

naXity – AI-Powered Urban Mobility & Smart City Intelligence Platform

Project Overview

naXity is a next-generation smart city platform that combines navigation, ride booking, food discovery, public transportation, emergency services, weather intelligence, community reporting, and AI-powered city insights into a single ecosystem.

The platform aims to solve everyday urban mobility challenges by providing real-time information, intelligent recommendations, and seamless access to city services.

Objectives

- Simplify urban transportation
 - Improve city navigation experience
 - Enhance public safety
 - Promote community participation
 - Provide AI-driven travel recommendations
 - Enable smart city decision-making through analytics
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User Roles

User

- Search locations
- Navigate routes
- Book rides
- Discover restaurants
- Report incidents
- View weather information
- Plan trips

Driver

- Accept ride requests
- Manage trips
- View earnings
- Track navigation

Restaurant Owner

- Manage restaurant profile
- Update menus
- Handle reservations
- View customer analytics

City Official

- Monitor city incidents
- Analyze traffic reports
- Manage public alerts

Admin

- Manage users
 - Manage reports
 - View platform analytics
 - Moderate content
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Module 1: Navigation System

Route Search

- Search destinations
- Location autocomplete
- Nearby place suggestions

Turn-by-Turn Navigation

- Voice navigation
- Visual route guidance
- Lane recommendations

Distance Calculation

- Travel distance
- Estimated travel time

Alternative Routes

- Multiple route options

- Fastest route
- Shortest route

Live ETA

- Dynamic arrival time updates
- Traffic-aware estimations

Traffic-Aware Routing

- Real-time traffic monitoring
 - Congestion avoidance
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Module 2: Ride Booking System

Ride Booking

- Book instant rides
- Schedule future rides

Fare Estimation

- Distance-based fare
- Dynamic pricing

Driver Dashboard

- Ride requests
- Earnings tracking
- Trip history

Driver Tracking

- Real-time driver location
- Route visualization

Ride History

- Previous rides
- Payment records

Ratings & Reviews

- Driver ratings
- Passenger ratings

Module 3: Food Discovery Platform

Restaurant Listings

- Nearby restaurants
- Featured restaurants

Search & Filters

- Cuisine filters
- Budget filters
- Rating filters

Ratings & Reviews

- User reviews
- Star ratings

Menu Browsing

- Restaurant menus
- Dish details

Table Reservations

- Reserve tables
 - Reservation management
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Module 4: Public Transport System

Bus Tracking

- Live bus locations
- Estimated arrival times

Metro Tracking

- Train locations
- Metro schedules

Route Suggestions

- Best public transport routes

Transit Timetable

- Bus schedules
 - Metro schedules
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Module 5: Weather Intelligence

Live Weather

- Temperature
- Humidity
- Wind Speed
- Pressure
- UV Index
- Visibility

Hourly Forecast

- Next 24-hour forecast

7-Day Forecast

- Weekly weather outlook

Air Quality Monitoring

- AQI Score
- PM2.5
- PM10

Weather Alerts

- Heavy Rain Alerts
- Heatwave Alerts
- Storm Warnings
- Fog Warnings

Weather-Based Routing

- Flood-prone route detection
 - Rain-safe route suggestions
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Module 6: Emergency & Safety System

Emergency SOS

- One-click emergency button
- Share live location

Emergency Contacts

- Notify family members
- Notify friends

Nearby Emergency Services

- Hospitals
- Police Stations
- Fire Stations

Safety Alerts

- Accident warnings
 - High-risk zones
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Module 7: Community Reporting System

Incident Reporting

Users can report:

- Accidents
- Roadblocks
- Potholes
- Waterlogging
- Construction Zones
- Traffic Jams

Upvote System

- Community validation
- Report prioritization

Live Community Feed

- Real-time city reports

City Leaderboard

- Top contributors
 - Citizen engagement
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Module 8: Smart City Services

Parking Finder

- Nearby parking spaces
- Live parking availability

EV Charging Stations

- Locate charging points
- Availability status

Petrol Pump Finder

- Nearby fuel stations
- Fuel price comparison

Public Washroom Finder

ATM Finder

Pharmacy Finder

Hospital Finder

Module 9: AI-Powered Features

AI Route Optimizer

Suggest:

- Fastest Route
- Cheapest Route
- Safest Route
- Fuel Efficient Route

AI Traffic Forecast

Predict traffic:

- 30 Minutes Ahead
- 60 Minutes Ahead

AI Crowd Prediction

Predict crowd density at:

- Markets
- Tourist Attractions
- Restaurants
- Events

AI Parking Recommendation

Suggest:

- Nearest parking
- Cheapest parking
- Available parking

AI Travel Assistant

Natural Language Queries:

Examples:

"Plan a one-day trip in Prayagraj under ₹2000"

AI Generates:

- Places to Visit
- Route Plan
- Food Stops
- Estimated Budget

AI Weather Advisor

Suggest:

- Best time to travel
- Weather risk assessment

Module 10: Social Features

Friend Tracking

- Share live location
- Temporary location sharing

Group Trips

- Collaborative trip planning

Travel Recommendations

- Share favorite places

Activity Feed

- Recent community activity
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Module 11: Analytics Dashboard

User Dashboard

Travel Statistics

- Trips Completed
- Distance Travelled
- Time Saved

Environmental Impact

- Carbon Saved
- Fuel Saved
- Eco Score

Personal Insights

- Most visited locations
- Travel habits

Admin Dashboard

Traffic Analytics

- Traffic heatmaps
- Peak traffic zones

Incident Analytics

- Report trends
- Accident hotspots

User Analytics

- Active users

- Growth metrics

Ride Analytics

- Total rides
 - Revenue reports
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Module 12: Smart Search System

Users can search cities such as:

"Prayagraj"

SmartCityX displays:

City Overview

- Population
- Weather
- AQI
- Traffic Status

Transportation

- Routes
- Public Transport
- Ride Availability

Food Discovery

- Restaurants
- Cafes
- Popular Foods

Essential Services

- Hospitals
- Pharmacies
- ATMs
- Petrol Pumps

Community Reports

- Active incidents

- Local alerts

AI Insights

- Best travel time
 - Traffic forecast
 - Crowd forecast
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Authentication & Authorization

Authentication

- Clerk/Auth.js
- Google Login
- Email Login

Role-Based Access Control

Roles:

- User
 - Driver
 - Restaurant Owner
 - City Official
 - Admin
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Notification System

Push Notifications

- Traffic Alerts
- Ride Updates
- Weather Alerts

Email Notifications

- Booking confirmations
- Incident reports

SMS Alerts

- Emergency notifications

Payment System

Razorpay Integration

Supports:

- Ride Payments
- Restaurant Reservations
- Premium Features

File Management

Cloudinary Integration

Store:

- User Photos
- Restaurant Images
- Incident Images
- Verification Documents

Database Design

Core Entities

- Users
- Roles
- Permissions
- Drivers
- Vehicles
- Trips
- Locations
- Routes
- Restaurants
- Menus
- Reservations

- Reviews
- Reports
- Incidents
- Parking
- EV Stations
- Weather
- Notifications
- Payments
- Analytics

Estimated:

25–30+ relational tables

Non-Functional Requirements

Performance

- Response Time < 2 Seconds
- Real-Time Updates

Scalability

- Microservice Ready
- Horizontal Scaling

Security

- JWT Authentication
- RBAC
- HTTPS
- Input Validation

Reliability

- Error Tracking
 - Monitoring
 - Logging
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Technology Stack

Frontend

- Next.js
- TypeScript
- Tailwind CSS

Backend

- Node.js
- Express.js

Database

- PostgreSQL
- PostGIS

Real-Time

- Socket.IO

Authentication

- Clerk / Auth.js

Cache

- Redis

File Storage

- Cloudinary

Payments

- Razorpay

Maps & Location

- Google Maps API

AI Services

- OpenAI / Gemini APIs

Monitoring

- Sentry

Logging

- Winston

Deployment

- Docker
 - AWS
 - Vercel
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Project Vision

naXity aims to become a unified platform that enables citizens to navigate cities smarter, travel safer, discover places efficiently, and actively contribute to building intelligent urban ecosystems through AI, real-time data, and community collaboration.